

Chapter 188

Delivering DCP and IMF

For projects requiring Digital Cinema Package (DCP) or Interoperable Master Format (IMF) mastering for digital cinema or broadcast distribution, DaVinci Resolve allows native DCP and IMF encoding and decoding for the creation and playback of unencrypted DCP and IMF deliverables, or it can be integrated with Fraunhofer's easyDCP application in order to master fully encrypted DCP files, play them back for testing, and generate Key Delivery Messages (KDMs) for theatrical distribution, all directly within DaVinci Resolve.

This means you can encode a DCP or IMF master straight from your program's source media, all within the 32-bit floating point image processing pipeline of DaVinci Resolve, for the highest possible quality result.

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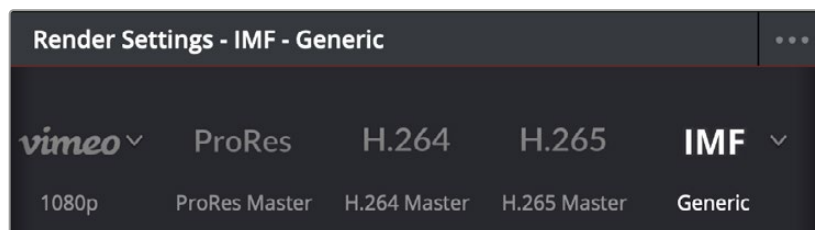
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Native IMF Encoding and Decoding (Studio Version Only)

The Format dropdown in the Video panel of the Render Settings now has a native IMF option that lets you export to the SMPTE ST.2067 Interoperable Master Format (IMF) for tapeless deliverables to networks and distributors. No additional licenses or plugins are required to output to IMF.

The IMF format supports multiple tracks of video, multiple tracks of audio, and multiple subtitle and closed caption tracks, all of which are meant to accommodate multiple output formats and languages from a single deliverable. As of DaVinci Resolve 16, IMF export also supports exporting IMF packages that use ST.2098 and Dolby immersive audio via selected Main busses. All of this is done by wrapping a timeline's different video and audio tracks (media essences) and subtitle tracks (data essences) into a "composition" within the Material eXchange Format (MXF).

Additionally, a dropdown menu to the right of this preset provides options for Generic, 20th Century Fox, and Netflix versions of this preset.



The IMF Generic preset in the Render Settings has options for different resolutions of output

When IMF is selected from the Format dropdown, the Codec drop-down menu presents options for Kakadu or EasyDCP encoding, with Kakadu being the method that's included with DaVinci Resolve Studio. A Type dropdown lets you choose what kind of JPEG2000 output you want, with options including RGB, YUV, and Dolby Vision. Additional parameters include:

- **Package Type:** Defaults to App2 Extended (App2e), for encoding JPEG 2000 up to 4K.
- **Bit Depth:** The bit depth of the encoded IMF video.
- **Encoding Profile:** A dropdown that lets you choose among Auto, IMF, and Broadcast.
- **Encoding Level:** Provides different choices based on what is selected in Encoding Profile.
- **Maximum bit rate:** Lets you choose how much to compress the result.
- **Lossless Compression:** Lets you choose to encode using lossless compression.
- **Slope-Rate Control:** A checkbox lets you specify lossless compression.
- **QStep:** (DCP, IMF) Lets you choose either automatic or manually specified DCP quantization levels at which to compress the video signal when using the Kakadu JPEG 2000 encoder.

Render settings in the Export Video section for the IMF format

A separate group of parameters named Composition Settings, underneath the Advanced Settings, lets you add metadata to your IMF package, including:

- **Composition name:** The name of the exported composition.
- **Issuer:** The organization providing the composition.
- **Use current date:** A checkbox that lets the current date be used as the Issue date automatically.
- **Issue date:** The date the composition is issued.
- **Content kind:** A dropdown provides a list of acceptable choices for defining the content.
- **Content version label:** Meant to identify the version of the content being provided.
- **Annotate xml using composition name:** Auto-populates Asset Map, Composition Playlist, and Packing List with data from the project. Otherwise these three fields are manually editable.
- **Annotate media using filename:** Auto-populates Main Video Track and Audio Track 1 with data from the project. Otherwise these three fields are manually editable.

Parameters for adding composition metadata

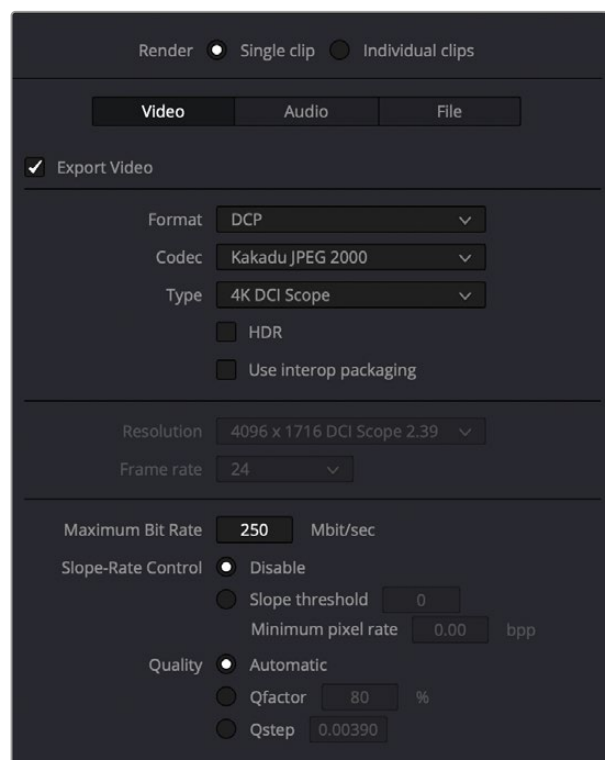
Native Unencrypted DCP Encoding and Decoding (Studio Version Only)

DaVinci Resolve also has native DCP encoding and decoding support built-in, for unencrypted DCP files only. That means that you can output and import (for test playback) unencrypted DCP files without needing to purchase a license of EasyDCP. If you have a license, a setting in the Configuration panel of the System Preferences enables you to choose whether to use EasyDCP (for creating encrypted DCP output), or the native DaVinci Resolve encoding.

Native DCP Encoding Parameters

When you choose DCP from the Format dropdown menu, the following additional parameters are exposed:

- **HDR:** (DCP, IMF) Specifies the package as having HDR content.
- **Use interop packaging:** (DCP only, located under Type parameter) Lets you create an Interop DCP package, based on an earlier standard of DCP delivery that is not forward compatible with SMPTE DCP packages.
- **Maximum bit rate:** (DCP, IMF) Lets you choose how much to compress the result.
- **Lossless Compression:** (IMF) Lets you choose to encode using lossless compression.
- **Slope-Rate Control:** (DCP, IMF) A checkbox lets you specify lossless compression.
- **Quality:** (DCP, IMF) Lets you choose either automatic or manually specified DCP quantization levels at which to compress the video signal when using the Kakadu JPEG 2000 encoder.



Native DCP settings in DaVinci Resolve

If you've selected DCP from the Format dropdown menu, a Composition Settings group appears with the following parameters when you click the disclosure control, which let you populate standard DCP composition metadata:

- **Composition name:** The name of the exported composition. DCPs use specific naming conventions for the composition name that include metadata about the file itself for DCP projectors and playback equipment. DaVinci Resolve has a tool called the Composition Name Generator to generate these names properly for you; it is accessed by pressing the “Edit” button next to this field. Simply fill out the fields and press OK, and DaVinci Resolve will rename your composition in line with these standards.

The screenshot shows the 'Composition Name Generator' dialog box. At the top, the generated name is displayed: `JohnsFilm_ADV-2D-DVis_F-185_EN-TH_US-13_51_2K_NULL_NUL_SMPTE`. Below this, there are several sections of controls:

- Film Title:** A text field containing 'JohnsFilm'.
- Content Type:** A dropdown menu set to 'ADV (A...ement)'.
- Content Version:** A text field set to '0'.
- Content Modifiers:** A group of checkboxes including 'Temp', 'Pre', 'RedBand', 'Chain', '3D', '2D' (checked), 'Mastered Luminance', 'Frame Rate' (set to 24), 'Dolby Vision' (checked), and 'Eclair Color'.
- Custom Modifiers:** A text field.
- Projector Aspect Ratio:** Radio buttons for 'None', 'Flat' (selected), 'Scope', and 'Full'. A checkbox for 'Use encode aspect ratio' is checked.
- Audio Language:** A dropdown menu set to 'EN (ENGLISH)'.
- Subtitle Language:** A dropdown menu set to 'TH (THAI)'.
- Territory:** A dropdown menu set to 'US (UN...TATES)'.
- Rating:** A dropdown menu set to '13 (PG-13)'.
- Audio Type:** A dropdown menu set to '51 (5.1 Surround)'.
- Resolution:** A dropdown menu set to '2K (2048 x 1080)'.
- Studio:** A dropdown menu set to 'NULL (S...cified)'.
- Date:** A checkbox for 'Include date' is checked, and a date field shows '20191204'.
- Facility:** A dropdown menu set to 'NUL (Unspecified)'.
- Standard:** Radio buttons for 'None', 'IOP', and 'SMPTE' (selected).
- Package Type:** Radio buttons for 'None' (selected), 'OV', and 'VF'.

At the bottom right, there are three buttons: 'Reset', 'Cancel', and 'OK'.

The Composition Name Generator will pass a standards compliant name for you to the Composition Name field.

- **Issuer:** The organization providing the composition.
- **Use current date:** A checkbox that lets the current date be used as the Issue date automatically.
- **Issue date:** The date the composition is issued.
- **Content kind:** A dropdown provides a list of acceptable choices for defining the content.
- **Content version label:** Meant to identify the version of the content being provided.
- **Annotate xml using composition name:** Auto-populates Asset Map, Composition Playlist, and Packing List with data from the project. Otherwise these three fields are manually editable.
- **Annotate reel index as suffix:** Auto-populates Reel Annotation with data from the project. Otherwise this is manually editable.
- **Annotate media using filename:** Auto-populates Main Video Track and Audio Track 1 with data from the project. Otherwise these three fields are manually editable.

Parameters for adding composition metadata

Rendering IMF Segments and DCP Reels

DaVinci Resolve supports splitting IMF and DCP projects into separate segments and reels, in addition to rendering media as a single file. This can be useful for breaking up your timeline to fit within allowed file sizes on legacy file systems, separating marketing and studio assets from the final film, or just being able to replace certain sections of the film without having to re-encode the entire file.

To Render an IMF Segment or DCP Reel:

- 1 Choose IMF or DCP from the format settings in the Video tab of the Render Settings panel.
- 2 Navigate to the “Segment list” or “Reel list” section of the composition settings.
- 3 Choose from the following options:

Single segment/reel: Encodes the Timeline into a single file (default setting).

Regular intervals of: Encodes the Timeline into multiple segments/reels, each one the duration of the value set in the “mins” field.

Align to clips on the first video track: Encodes the Timeline into multiple segments/reels; each individual clip on the V1 track of the Timeline becomes its own separate file.

Keep audio as single segment (IMF only): Select this checkbox to keep the audio portion of the IMF as a single file, regardless of the segment options selected above.

- 4 Press the Add to Render Queue button.

IMF Segment options

Creating DCP/IMF Supplemental Packages

Once created, DaVinci Resolve has the ability to reimport a DCP or IMF so that you can overwrite parts that need to be updated with new media, in order to export a “Supplemental Package,” which is effectively a new version of the program that combines the new overwritten parts of the program with the old version, such that you can deliver just the changes.

NOTE: Supplemental packages are only supported using the Kakadu encoder and decoder; this is not compatible with DCP or IMF packages created using EasyDCP. To avoid issues, disable “Use easyDCP decoder” in the Decode Options panel of the DaVinci Resolve System Preferences.

Importing a DCP or IMF Into a Timeline

- 1 Using the Media Storage browser in the Media page, find and select the DCP or IMF, and check the header of the Metadata Editor to verify that your media is suitable for creating a supplemental package. Supported IMF profiles will be displayed in the Metadata Viewer.

Metadata	Media Storage	...	⇅
mer_shrt_2398_pqp3d65...0_dovi_imf_20170913_1 00:12:02:17			
/Users/jackh/Public/MediaPool/IMF			
IMF Dolby Vision M6S3	23.976 fps	3840 x 2160	
Linear PCM	48000 Hz	8 Ch	

The header in the Metadata Editor showing an IMF that’s compatible with the creation of a supplemental package

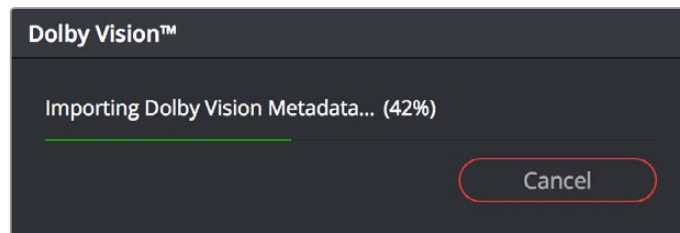
- 2 Create a new project and add the DCP/IMF package you need to modify to the Media Pool. If a dialog appears asking if you want to change your timeline frame rate to match the incoming media, click Change to make your project match the media.
- 3 Create a timeline from the composition playlist (XML) within the imported DCP or IMF by right-clicking the imported package in the media pool and choosing “Create New Timeline Using Composition Playlist” from the contextual menu.



Right-clicking an imported IMF or DCP clip in the Media Pool reveals a command to make a new timeline using the composition playlist in the contextual menu

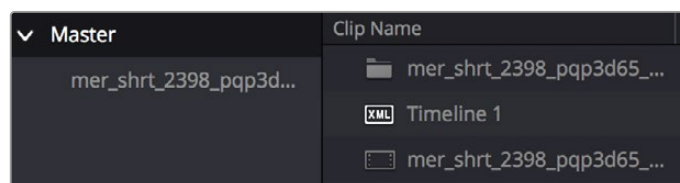
- 4 The New Timeline dialog has an “Import Dolby Vision Project Settings” checkbox. When it’s turned on, clicking Create will do the following:
 - a) Dolby Vision will be enabled in the Color Management panel of the Project Settings, and the Mastering Display menu will be set to match that of the IMF package.
 - b) If Resolve Color Management (RCM) is not active, the Timeline Color Space will be set to match the Dolby Vision metadata. However, if RCM is already enabled, the user must manually set this by turning on “Use Separate Color Space and Gamma,” and changing the Timeline settings to P3-D65” and “ST.2084” respectively.

- 5 Creating the Timeline will import Dolby Vision metadata (if applicable). This will allow a Tone Mapping preview to be seen on the Color page that uses the original metadata.



Dolby Vision metadata will be imported if present when importing an IMF

Once import is complete, all video and audio clips from the DCP or IMF appear within a new bin with the name of the package. The resulting timeline will be identified via its icon as a DCP/IMF timeline.

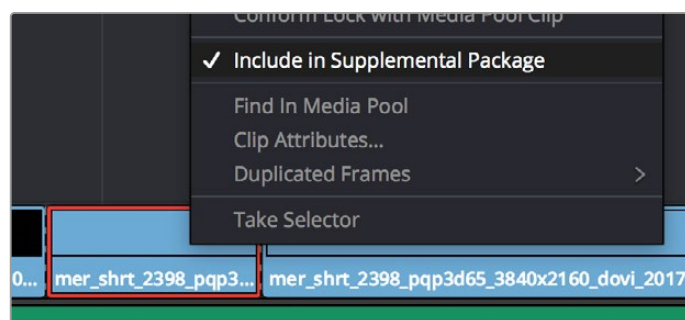


The imported media and timeline when you import an IMF

Editing the Resulting Timeline

At this point, you can edit the program in the Timeline as necessary.

- You can overwrite sections of the Timeline with new clips. All modifications will be automatically included into the supplemental package.
- You can use the Blade tool or Insert Edit command to cut sections of the existing program to which you want to add Fusion effects, audio grading, or color correction. When you do this, you must right click that section and choose “Include in Supplemental Package” to make sure it exports correctly.



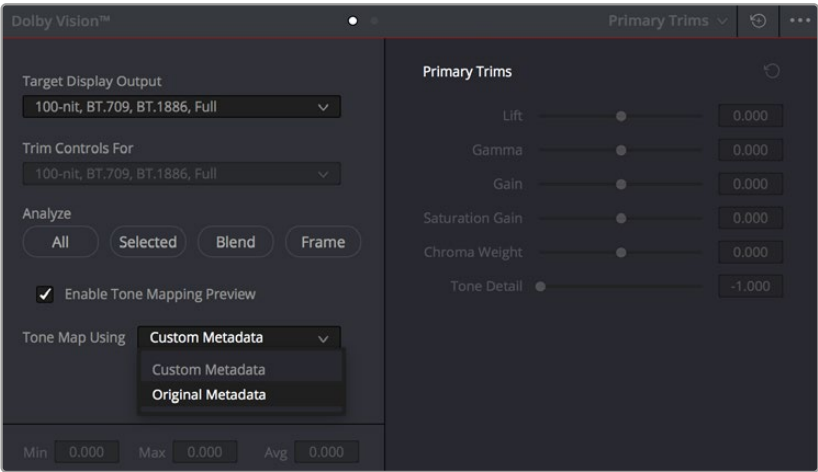
Right-clicking a section of the program and choosing Include In Supplemental Package

For IMF Dolby Vision packages, please ensure all modifications are on the first video track (V1).

NOTE: If RCM is being used, please ensure the input color space and gamma of the inserts are correct.

Dolby Vision Metadata

The Dolby Vision metadata from the imported DCP/IMF file can be reused by selecting “Original Metadata” from the Tone Map Using dropdown menu of the Dolby Vision palette in the Color Page

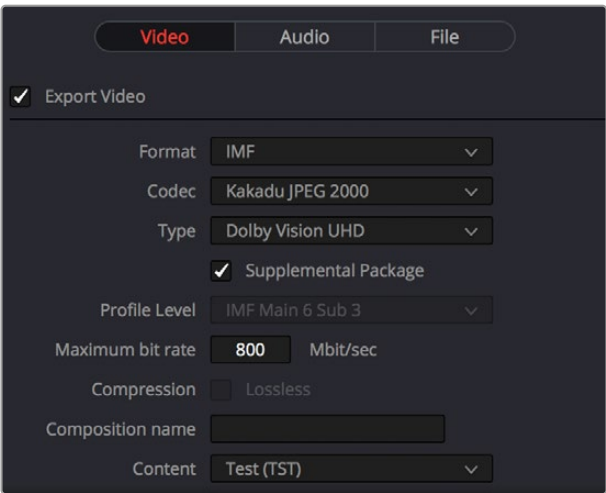


Setting a clip to Original Metadata in the Color Page

Alternatively, this metadata can be imported separately from an existing XML via the “Import Metadata from XML” command in the option menu of the Dolby Vision palette in the Color page. When successful, “Imported Metadata” will be enabled.

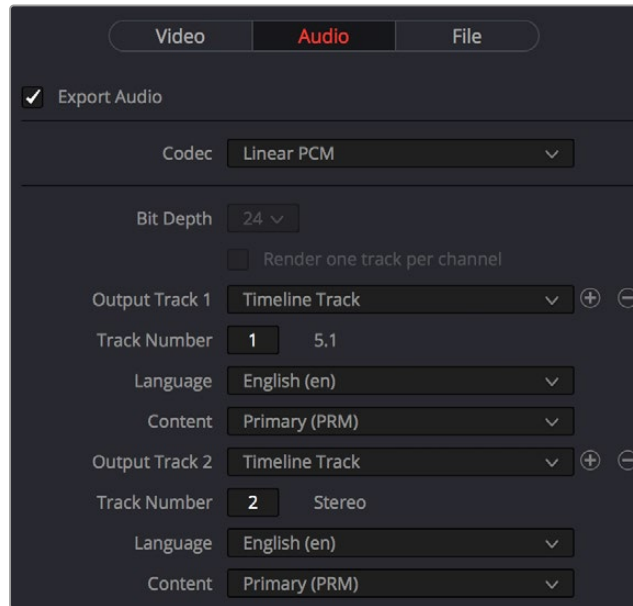
Exporting

You can export a supplemental package by turning on “Supplemental Package” in the the video panel of the Deliver page Render Settings list.



Setting an export to be a Supplemental Package

The codec type and profile will be automatically selected to match the original version of the DCP/IMF package, and the audio tracks are set to match the Timeline tracks. Please ensure the rest of these audio settings are matched to the original version, since they start out set to the default values.



Setting the audio settings of a Supplemental Package export

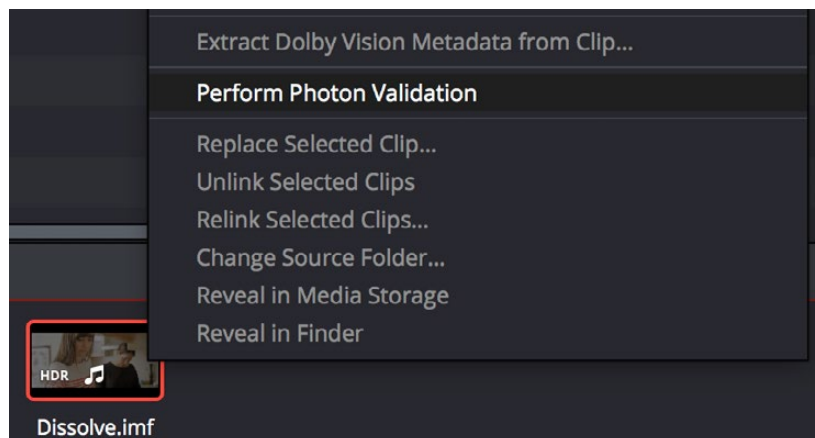
Photon Validation of IMF Packages

Photon is Netflix's validation software for IMF App2/App2e packages. The option for using Photon validation will only be shown on Resolve Studio with JDK/JRE version 1.8 and above installed, which is available at <https://github.com/Netflix/photon>.

NOTE: Please disable "Use easyDCP decoder" from "Preference" as there can be issues decoding IMF packages without an easyDCP license.

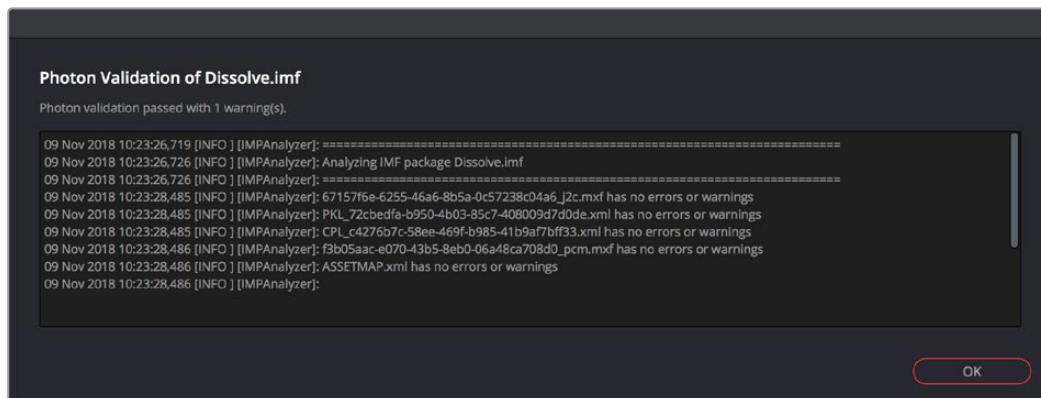
Validating in the Media Pool

An existing IMF package can be validated with Photon by importing it into the Media Pool, then right-clicking it and choosing "Perform Photon Validation" from the context menu.



Validating an IMF in the Media Pool

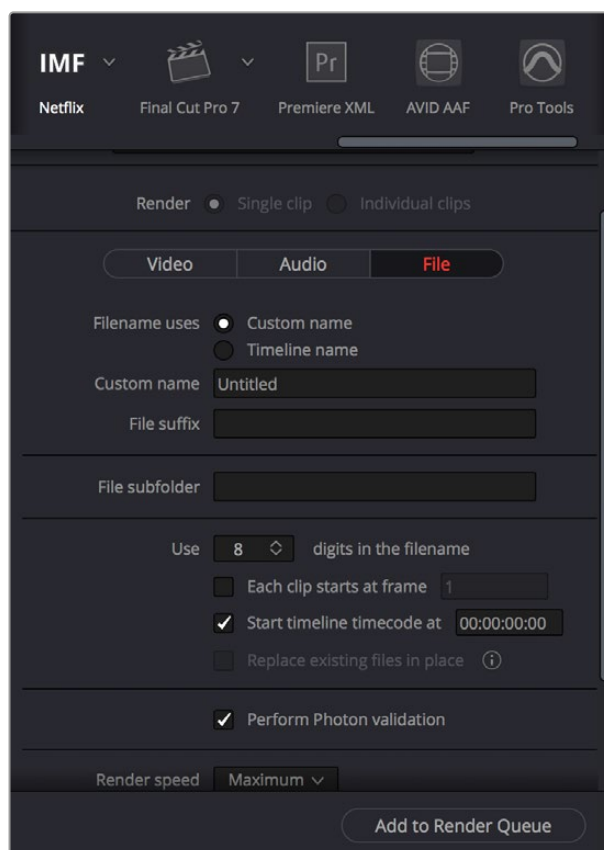
A report dialog will be shown when the validation is completed.



A validation report

Validating on Export

Photon validation can be enabled in the File panel of the Deliver page Render Settings; choosing the "IMF Netflix" preset will also enable this option. When enabled, DaVinci Resolve will perform Photon validation after the IMF package is exported. The validation report will be saved to a text file in the IMF package folder, and a report dialog will be shown if there is any error.



Enabling Photon validation on export

Using and Licensing EasyDCP

Both DaVinci Resolve and DaVinci Resolve Studio include a demo version of easyDCP. Details of operation and restrictions of the demo version can be found later in this chapter. The fully functional version of easyDCP operates via licensing modules purchased from <http://www.easyDCP.com> (info@easyDCP.com) and every new DaVinci Resolve system (server) needs its own license and specific certificates for DCP and KDM generation and for playback of DCPs.

Requesting Your Server Certificate Set

For your DaVinci Resolve system to generate DCPs and KDMs you need to request a specific set of configuration files called the Server Certificate Set. To begin, first purchase your encoding, encryption, decoding, and decryption modules from easyDCP. They will provide a password to access your easyDCP account.

Then, from the DaVinci Resolve File menu, select easyDCP > Request Server Certificate Set. Fill in the detail listed on the request form and save the form to your desktop or somewhere it can be easily found. This html file can be emailed to info@easyDCP.com. After sending the html, a customized Server Certificate Set for your installation will be generated and made available for download in your easyDCP Website User Account.

The Server Certificate Set generated for your DaVinci Resolve will contain files based on your purchased modules and your specific DaVinci Resolve server hardware. The table below shows the modules and the licenses and certificates generated, followed by a brief description of each item.

	License	Server Certificate	Signer Certificate
DCP Encoder	X		X
DCP Encoder with Encryption	X	X	X
DCP Player	X		
DCP Player with Encryption	X	X	

License: The License is used to activate the purchased modules on a specific hardware server.

Server Certificate: Each DCP render (referred in the industry as an “Instance”) using encryption or decryption has an individual server certificate. This certificate is required to be able to receive Key Delivery Messages (KDMs), which unlock encrypted DCPs.

Signer Certificate: A Signer Certificate is used to sign certain files within a DCP package and/or Key Delivery Message (KDM) to verify which authority generated the DCP instance.

Importing Your Server Certificate Set

Once generated and downloaded to your DaVinci Resolve server, the Server Certificate Set needs to be imported into DaVinci Resolve.

To import your server certificate:

- 1 Choose File > easyDCP > Import License and Certificates.
- 2 Use the Import Server Certificate dialog to select the file, enter your Certificate Set password, then click Import.
- 3 To verify your easyDCP license and the Server Certificates, choose easyDCP > About easyDCP.

From this point onward, you can use the controls from within the Settings window, the Deliver page, and the File menu to master and play DCPs.

Limitations of the Demo Version of easyDCP

The demo version of the DCP encoder embeds visible DaVinci Resolve and easyDCP logo watermarks in the rendered Digital Cinema Package (DCP) images. The demo version does not include encryption so these DCPs can be used for screening in a digital equipped cinema. The demo version of the DCP playback module will play 15 seconds in full quality. After that playback quality reduces drastically. Furthermore, audio won't be rendered after 15 seconds of playback.

Switching Between Native DCP and EasyDCP Encoding

A checkbox in the Configuration panel of the System Preferences, "Use EasyDCP Encoder," lets you choose whether to use the native DCP/IMF encoding in DaVinci Resolve, or your licensed EasyDCP software. In either case, all set up happens from within the Deliver page of DaVinci Resolve.

EasyDCP Color Management

The Color Management panel of the Project Settings has a Timeline Colorspace dropdown menu that is enabled for EasyDCP encoding regardless of whether or not DaVinci Resolve Color Management is used for the current project (the same setting is used for both color management tasks). You should set this to the color space used by your current DaVinci Resolve timeline. If, for example, you are grading using a Rec. 709 monitor for television deliverables but also wish to make a DCP, select Rec. 709 Gamma 2.4 and DaVinci Resolve will render the DCP with the correct Rec. 709 to XYZ matrix.

EasyDCP Output in the Deliver Page

To master to a DCP in the Deliver page, use the following procedure, which walks you through all of the easyDCP settings that are available in the Render Settings list.

To master a DCP or IMF:

- 1 Set "Render timeline as" to Single clip.
- 2 Choose easyDCP from the Video Format dropdown.
- 3 Choose the appropriate option from the Codec dropdown that corresponds to the type (DCP or IMF) resolution (2K or 4K), and aspect ratio (native, scope, or flat) of your intended output.

- 4 Set the Composition Name. This field is intended to hold a standardized name for the DCP being encoded. You can either type a name into this field directly, or you can press the “...” button to open the easyDCP Composition Name Generator window. An editable Film Title field appears, along with a number of dropdown menus that let you select various DCP attributes such as content type, aspect ratio, language of audio and subtitles, and so forth. As you populate each attribute, the name being generated appears at the top of the window, and clicking OK copies the resulting Composition Name into the Composition Name field of the Render Settings.
 - 5 If necessary, set the desired “Maximum DCP bit rate” by either typing or dragging within the field (the range is 50 to 250 Mbit/sec). If you’re not sure what data rate to use, consult the client or distributor to whom you’re delivering the DCP.
 - 6 There are two DCP package types you can output, determined by the “Use Interop packaging” checkbox:
 - The standard package conforms to the “Interop” specifications for DCPs, which is turned on by default. With “Use Interop packaging” turned on, however, the frame rate of your output is limited to either 24fps or 48fps, so you need to make sure that your timeline conforms to these frame rates.
 - If you want to generate DCP packages with other frame rates to match your timeline, you need to turn “Use Interop packaging” off to generate a SMPTE-standard DCP. This supports additional frame rates including 25, 30, 50, and 60 fps. However, SMPTE-Standard-DCPs are not supported on all JPEG2000-based playback systems so it’s generally recommended to use the Interop standard unless you know the player supports the SMPTE-Standard DCPs.
 - 7 Turn on the “Encrypt package” checkbox to encode an encrypted DCP. This sets the encoder to generate a Digest containing the keys used during encryption. This Digest will allow you to play the resulting DCP on your system, and to generate KDMs to allow that DCP to be played on other servers.
- NOTE:** If you do not encrypt the DCP it can be played on any DCP player/decoder without restriction.
- 8 Set the Subtitles Path. If you have a properly formatted subtitle file, click the Browse button to link to it.
 - 9 If you’re including an audio mix in the DCP, go to the Audio section, turn on the Render audio checkbox, and choose the number of channels in the “Render channels of audio” dropdown menu that corresponds to the number of Audio Mixer output channels defined in the Edit page.
 - 10 Click the Browse button under the “Render to” field, and choose a location for the resulting DCP. Make sure you pick a drive with enough room for the estimated size of the final DCP.
 - 11 Choose all necessary options from the Output Options to ensure the quality you need.
 - 12 Click the Add Job to Render Queue button, and then click Start Render to create your DCP. A DCP will be created and placed at the location you chose, ready for playback or delivery.

KDM Generation and Management

Key Delivery Messages (KDMs) are required to allow an encrypted DCP play on a designated projector at a particular theater at a specified time. DaVinci Resolve is capable of creating KDMs, which is convenient for exporting KDMs for select screenings, but commercial distributors may require thousands of KDMs. Fortunately, easyDCP allows you to use external Distribution KDM (DKDM) utilities to generate KDMs for your clients, so you don’t have to tie up your DaVinci Resolve workstation with this task.

Publishing Your Encrypted Digital Cinema Package

While you can play your encrypted DCP on the same DaVinci Resolve system that generated it, if you wish to publish the DCP so other players can decode and play you need to generate a KDM to send to the player. The user of the other player, or players, will need to generate a Server Certificate for each of their players and send this to you so when you generate the KDM it will be just for those players.

Select the DCP in the Media page Library. Right-click and select Generate KDMs. From the dropdown select the location of the Server Certificate file if the KDM is for one player, or folder for multiple players. Set the start and end dates that the KDM will be valid for, an output folder to place the KDM, and then select Generate.

You can now send your DCP and the KDMs to the player you authorized. The user there will import the KDM and the DCP will play between the start and end dates.

Playing Your Digital Cinema Package

To play a DCP you've output from DaVinci Resolve, use the Media page to add it to the Media Pool and edit it into a timeline like any other clip.

Decoding the JPEG2000 images embedded within the DCP in real time is computationally intensive. If your system is underpowered you can reduce the decoded resolution of the files by selecting Half or Quarter Resolution Decode from the File > easyDCP menu. A smaller, less bandwidth-intensive version of the JPEG2000 files will be decoded by discarding some levels of the wavelet stage inside the decoder, which will directly increase the playback performance.

Playing Third-Party Digital Cinema Packages

To play a non-encrypted DCP simply select the DCP in the Media page like any other clip. To play an encrypted DCP from a third party you first must publish your Server Certificate. They use the certificate to generate KDMs for their DCP to play on your DaVinci Resolve system. From the File menu choose easyDCP > Export Server Certificate, and on the dropdown menu choose a location to save the file. Send this to the third party for KDM generation.

When you receive a KDM or a Digest for an encrypted DCP you must first import the file into your DaVinci Resolve system. Choose File > easyDCP menu > Import KDM/Digest, and then select the file. Then simply select the encrypted DCP in the Media Page Library to play.